



COURSE DESCRIPTION FORM

INSTITUTION _____ **FAST-NUCES** _____
BS Computer Science _____

PROGRAM (S) TO BE EVALUATED

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	CS-4045
Course Title	Deep Learning for Perception
Credit Hours	3
Prerequisites by Course(s) and Topics	N/A
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	1. Assignments + Homework (10 %) 2. Quizzes (5 %) 3. Midterms (30 %) 4. Project (15 %) 5. Final Exam (40 %)
Course Coordinator	Saif ul Islam
URL (if any)	https://classroom.google.com
Current Catalog Description	
Textbook (or Laboratory Manual for Laboratory Courses)	1. Deep Learning by Yoshua Bengio
Reference Material	1. Deep Learning for Computer Vision: Expert Techniques to Train Advanced Neural Networks Using TensorFlow and Keras



Course Goals Mapping of CLOs & PLOs	CLO#	CLO description	BT Domain/ BT Level	PLO #
	CLO 1	Explain deep learning architectures and the mathematical models behind them.	C2 (Analyzing)	PLO 3
	CLO 2	Apply computer vision techniques and methods to perception tasks.	C3 (Applying)	PLO 5
	CLO 3	Produce a research paper through a structured project on deep learning for Perception	C3 (Applying)	PLO 4
Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Week	Contents		
	1	Introduction and transition Machine Learning Recap		
	2	Multilayer Perceptron, Backpropagation Introduction to CNNs		
	3-4	Introduction to CNNs and CNN Architectures		
	5	Optimizers, Loss Functions		
	6	Image Classification, Transfer Learning		
	7	Object Detection YOLO Framework		
	8	Image Segmentation U-Net Architecture		
	9	Federated Learning		
	10	Transformers in Vision (ViT)		
	11	Generative Modeling DCGAN, CycleGan, Pix2Pix		
	12	Perception with Multimodal Data		
	13-14	Project and Presentation		
Laboratory Projects/Experiments Done in the Course				



Programming Assignments Done in the Course				
Class Time Spent (in credit hours)		Problem Analysis	Solution Design	Social and Ethical Issues
	20%	40%	35%	5%
Oral and Written Communications	Every student is required to submit a Project (In Group of 3) and appear for viva.			

Instructor Name _Muhammad Saif ul Islam

Instructor Signature _____

Date _26-Aug-2025_